The weak error is usually defined with reference to suitable test functions (observables) $\phi(x)$ by

$$\mathrm{Err}^{\mathrm{weak}}_{h,\tau,\phi} = \max_{n=0,1,2,\ldots,\nu} |\mathbb{E}\phi(\mathrm{X}_n) - \mathbb{E}\phi(\mathrm{X}(nh))| \,,$$

and we say that a method has weak order r if $\mathrm{Err}^{\mathrm{weak}}_{h,\tau,\phi} < D(\tau)h^r$ for some $D(\tau)$, for all members of the chosen class of observables. It is easy to see that methods with high strong order give us high weak order too, but the converse is generally false.

As the focus of this chapter is on the computation of averages, the concept of weak order is more appropriate in evaluating methods.[1] Yet, even the weak order of convergence may not be suitable in molecular dynamics applications, or rather, we are interested in the very long term behavior of the weak error which is governed by the coefficient $D(\tau)$ as $\tau \to \infty$ [270, 356].

The standard approach to analyze the weak error would be based on computing a recurrence relation for $|\mathbb{E}\phi(\mathrm{X}_n) - \mathbb{E}\phi(\mathrm{X}(nh))|$ by use of the SDE and the methods of stochastic analysis. This approach mirrors the forward convergence analysis of ODE methods discussed in Chap. 2. It tends, however, to overestimate the magnitude of the quantity $D(\tau)$. In what follows, we show that, when the emphasis is on stationary averages, a more direct approach can be used to gain insight into the error based on an asymptotic expansion. In some cases, we are able to resolve explicitly the leading error term of computed averages in this long-time limit. The approach used here is analogous in many ways to the backward error analysis of deterministic problems presented in Chap. 3.

The *sampling error* is an additional error term that results from the finiteness of our trajectory: both in our inability to sample over an infinite time (for Langevin dynamics, we recover our averages only in the limit of $\tau \to \infty$) and our generation of only a finite number of points (averages are calculated from our discretized trajectory of ν steps). In large, high-dimensional systems (such as many relevant biomolecular applications) often the computational resource required to sample long enough in time to recover good averages is not available—the challenge then can be viewed as an *exploration* problem. Obtaining a complete sampling of the phase space in a system with rare transitions is extremely difficult, and is the most important challenge in molecular simulation [401].

However, we emphasize in this chapter that the discretization error (or *perfect sampling bias*) is also a very important contribution to the overall error, and in some cases may dominate the sampling error (for example, in systems where local resolution of the measure is more important than the global search). This discretization bias cannot be removed by considering a longer simulation time, so in some sense it is akin to a modelling error. Indeed as computing power increases and new algorithms are utilized, it is likely that trajectory generation will always be

[1] In practice all the methods presented here are convergent in the strong sense, but their strong convergence orders may be low, typically $r = 1/2$.